Exercise 21

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Introduction to Electrodynamics

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Solutions

Verified

Solution A   Solution B

4.8 (6 ratings)   Answered 2 years ago

Step 1

1 of 3

Recall that the electric field outside a uniformly charged solid sphere is exactly the same as if the charge were all at a point in the center of the sphere:

$$\mathbf{E}_{\text{outside}} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{\mathbf{r}}$$

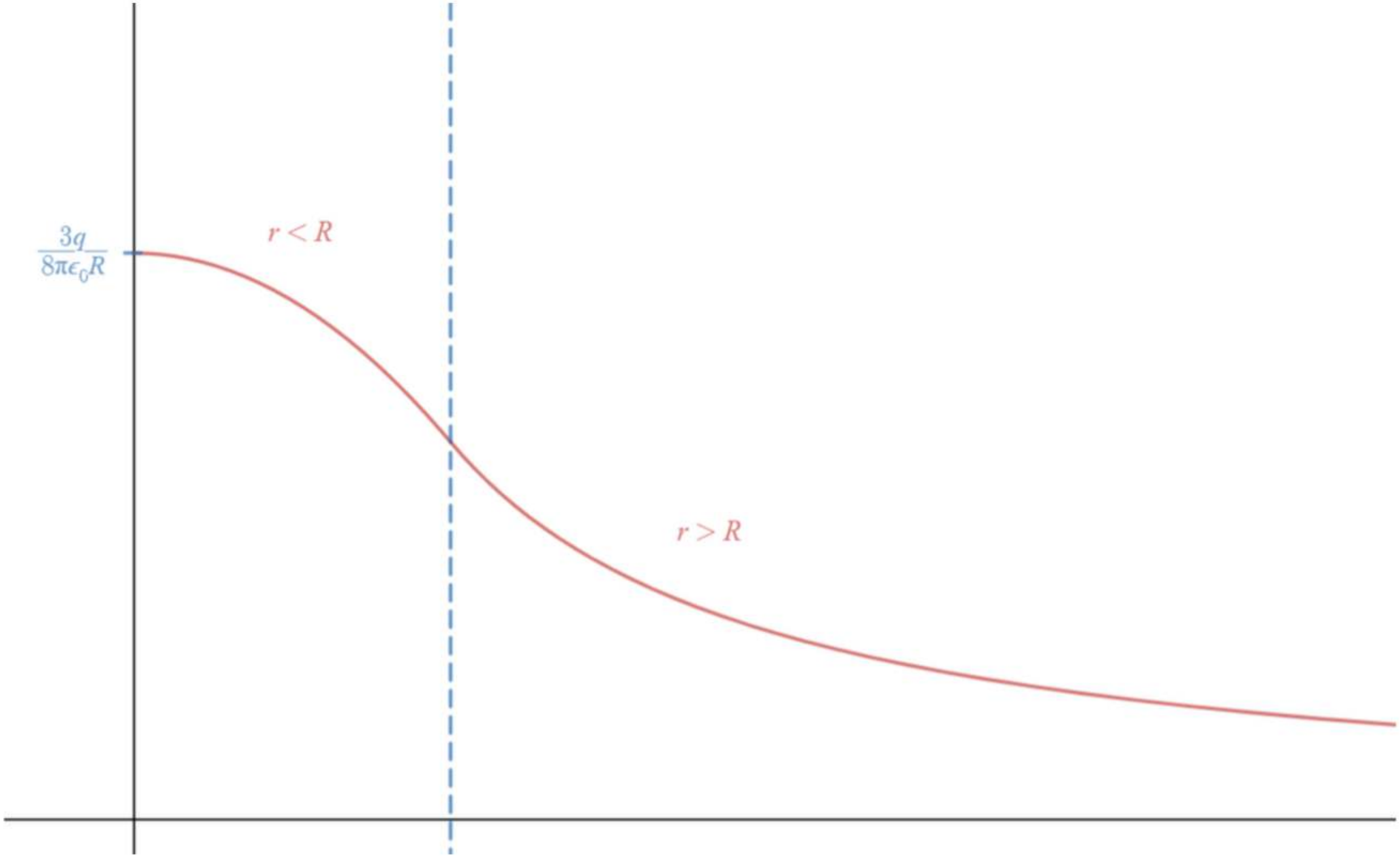
Inside the sphere, the electric field also acts like a point charge, but only for the proportion of the charge further inside than the point r:

$$\mathbf{E}_{\text{inside}} = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^2}\frac{r}{R}\hat{\mathbf{r}}$$

To find the potential, we integrate the electric field on a path from infinity (where of course, we take the direct path so that we can write the it as a 1D integral):

Step 2

Sketching V with some typical values for R and q, we get something like:



Result

$$V(r) = \begin{cases} \dfrac{q}{4\pi\epsilon_0}\left[\dfrac{1}{R} - \dfrac{r^2 - R^2}{2R^3}\right], & r < R \\ \dfrac{q}{4\pi\epsilon_0}\dfrac{1}{r}, & r > R \end{cases}$$

Exercise 20

Rate this solution

Exercise 22